



DATA COLLECTION STRATEGY & DATA QUALITY ASSESSMENT

Submitted By

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Domain

Apparel & Textiles – Data Analytics & Machine Learning

Project: Apparel Demand Forecasting and Overstock/Understock Risk Identification

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INTRODUCTION

The Apparel and Textiles industry generates data from sales transactions, product catalogues, inventory systems, customer behaviour, market reports, production statistics and digital trends. For a machine learning project, collecting data is only the first step; the data must also be relevant, reliable, consistent and sufficiently complete. Poor-quality data can lead to incorrect demand forecasts and weak identification of products at risk of overstock or understock.

This Week 2 task develops a practical data collection and data quality assessment strategy for the proposed project, “Apparel Demand Forecasting and Overstock/Understock Risk Identification.” The strategy focuses mainly on public and legally accessible data sources that can support future analysis. It considers government databases, open retail datasets, industry reports, online trend information and market research sources. It also defines extraction methods and a structured quality framework so that the collected data can later be prepared for machine learning.

1. Project Problem and Data Requirements

- The project aims to forecast apparel-related demand and identify products that may experience overstock or understock. The central analytical requirement is therefore a dataset containing historical product demand over time, together with useful product and transaction attributes.
- Potential variables include product ID, product category, product type, quantity sold, unit price, transaction date, sales value, country or region, customer segment, inventory level where available, and promotional or seasonal indicators. Additional external variables such as holidays, market trends and textile production statistics may improve the interpretation of demand patterns.
- The target data should ideally contain enough historical observations to identify weekly or monthly seasonality. Data should also have clear field definitions and consistent units. Where inventory information is unavailable in public datasets, demand can first be modelled from sales quantities and later extended with inventory data from an appropriate source.

2. Proposed Data Source Evaluation

- Each source will be evaluated before inclusion. The evaluation will consider authority, relevance, coverage, accessibility, documentation, licensing, update frequency and expected data quality.
- Primary transaction datasets are preferred for demand modelling because they directly represent observed purchases. Government and international datasets are useful for external market context and trend analysis. Industry reports can support interpretation but may not always be machine-readable. Social or search trends can provide timely signals but may contain sampling bias and should not be treated as direct evidence of actual sales.
- A source register will be maintained with fields such as source name, provider, URL, dataset description, date accessed, time coverage, geographic coverage, variables, licence, extraction method and known limitations. This creates traceability and makes the final analysis reproducible.

3. Data Extraction Methods

Different extraction techniques will be selected according to the structure and permission of each source.

- 1. Direct dataset download:** CSV, Excel or other open-data files can be downloaded when the provider offers them. This is simple, reproducible and generally preserves the published structure, but files may require manual updating.
- 2. APIs:** Where an official API is available, it can be used to retrieve structured data programmatically. APIs improve repeatability and can support scheduled collection, although authentication, rate limits, changing endpoints and documentation requirements may create limitations.
- 3. Web scraping:** Scraping may be considered only for publicly accessible information where the website's terms, robots rules and applicable policies permit it. Tools such as Python requests and BeautifulSoup can extract structured fields from suitable pages. Scraping can be affected by changing page layouts, missing fields, rate limits and anti-bot mechanisms, so it should not be the default method when an official dataset or API exists.
- 4. Manual structured collection:** For a small number of market indicators from reports, information can be entered into a controlled spreadsheet. This is practical for limited data but has a higher risk of transcription errors.
- 5. Surveys:** If original consumer information is required, a short voluntary survey can collect preferences or purchase-related opinions. Survey data would be supplementary because responses may be subjective and may not represent the entire market.

4. Data Collection Workflow

- The proposed workflow begins with defining the exact analytical questions and required variables. Next, candidate sources will be identified and documented. Each source will be checked for authority, licensing and relevance before extraction.
- After collection, files will be stored using a consistent naming convention and source metadata will be recorded. Raw data will be preserved separately from cleaned data so that transformations can be traced. The next stage is profiling, where column types, missing values, duplicate records, unique values, ranges and date coverage are examined.
- Cleaning will then address issues such as inconsistent category names, invalid dates, duplicated transactions, missing values and incorrect numeric formats. Finally, quality scores and validation results will be recorded. Only data that meets the project's minimum quality thresholds will be moved into the modelling dataset.

5. Data Quality Assessment Framework

- A structured quality framework will be used to assess every candidate dataset. The main dimensions are:
- Accuracy – Values should represent the underlying business event correctly. Examples include checking whether quantities and prices fall within reasonable ranges and whether calculated sales values agree with quantity $\times$ price where appropriate.
- Completeness – Important fields should have sufficient non-missing values. Missingness will be calculated for each variable and investigated before modelling.
- Consistency – The same product, category, date and unit conventions should be used across records. For example, category labels such as “T-shirt”, “T Shirt” and “Tshirts” should be standardised where they represent the same category.
- Timeliness – Data should be recent enough for the intended analytical question. Historical data remains useful for seasonality, while recent data is important for current trends.
- Validity – Values should follow expected formats and business rules. Dates must be valid, quantities should be numeric, and categorical fields should use accepted values.
- Uniqueness – Duplicate transactions or repeated records should be detected. Duplicate removal must be performed carefully because repeated-looking records may sometimes represent genuine transactions.
- Relevance – Variables should contribute to the project objective. Unrelated fields increase complexity without necessarily improving forecasting.
- Traceability – Every dataset should have a known source, collection date, transformation history and licence/usage information.

6.Data Quality Scoring

- To make source selection systematic, a simple quality score can be assigned to each dataset. Each dimension can be rated from 1 to 5. A possible weighted score is: Accuracy 20%, Completeness 15%, Consistency 15%, Timeliness 10%, Validity 10%, Uniqueness 10%, Relevance 15% and Traceability 5%.
- The final score can be calculated as the weighted average of the dimension scores. Datasets scoring 4.0 or above can be considered high quality for initial use, scores from 3.0 to below 4.0 can be considered usable after additional cleaning, and scores below 3.0 should be reviewed carefully or rejected. These thresholds are project-level guidelines rather than universal industry standards.
- Quality checks will also be performed after cleaning because transformations can introduce errors. A data-quality log will record the issue, affected column, number of records, correction applied and validation result.

7.Data Cleaning and Validation Plan

- The first cleaning stage will standardise column names and data types. Date fields will be converted into a consistent date format, numerical fields will be checked for non-numeric values, and categorical fields will be normalised.
- Missing values will be analysed rather than automatically replaced. If a critical target field such as sales quantity is missing, the record may need to be excluded from a modelling dataset. For less critical variables, an appropriate imputation or “Unknown” category may be considered depending on the variable.
- Outliers will be investigated using summary statistics and visual checks. An unusually high quantity may be a genuine wholesale transaction rather than an error, so domain context is necessary before removal. Duplicate detection will use relevant transaction identifiers and combinations of fields. Cross-field validation will check logical relationships such as positive quantities, non-negative prices where applicable, and chronological ordering.
- The cleaned dataset will then be split into raw, cleaned and modelling-ready versions. This separation helps preserve the original evidence and supports reproducibility.

8.Tools and Technologies

- Python will be used as the main data preparation and analysis language. Pandas can support tabular data loading, cleaning, missing-value analysis and transformation. NumPy can support numerical operations. Matplotlib can be used for data-quality visualisations such as missing-value summaries and distribution plots.
- Excel can be used for initial inspection and small manual datasets. Jupyter Notebook can document extraction, cleaning and quality checks in a reproducible form. Git and GitHub can be used to maintain project files, notebooks and documentation when required by the internship. APIs can be accessed through suitable Python libraries, while web extraction, if permitted, can use requests and BeautifulSoup.
- The tool choice will depend on the source format and permissions. Automated methods will be preferred for repeatable datasets, while manual collection will be limited to small contextual datasets.

9. Risks, Limitations and Ethical Considerations

- Public datasets may contain sampling bias, incomplete historical coverage, inconsistent definitions or outdated records. A retail dataset from one region may not represent the entire Indian apparel market.
- Fashion trends can also change rapidly, meaning historical patterns may not always continue.
- The project will respect dataset licences, website terms and applicable policies. Personal information will not be collected unnecessarily. If customer-level records are used, only fields necessary for the analysis should be retained, and personally identifying information should be excluded from the modelling dataset.
- Web scraping will not be used to bypass access controls or restrictions. Source attribution will be maintained, and downloaded datasets will be accompanied by their original source information. These practices help ensure that the data collection process is responsible, transparent and reproducible.

10. 30–35 Hour Work Plan

- The planned effort for Week 2 is approximately 34 hours.
- Hours 1–4: Review the project problem and define data requirements.
- Hours 5–9: Identify and compare public retail, government, international and market sources.
- Hours 10–14: Inspect dataset documentation, licensing and variable availability.
- Hours 15–19: Design extraction methods and collect representative datasets.
- Hours 20–24: Perform initial profiling for missing values, duplicates, formats and ranges.
- Hours 25–28: Build the data-quality framework and scoring method.
- Hours 29–31: Document cleaning and validation rules.
- Hours 32–34: Prepare the report, review limitations and organise the final submission.
- This schedule provides time for both practical data investigation and documentation rather than focusing only on downloading files.

Conclusion

- A reliable machine learning solution for the Apparel and Textiles industry depends on reliable data. This Week 2 strategy therefore treats data collection and data quality assessment as connected activities. Multiple public sources are considered so that transaction-level sales information can be combined, where appropriate, with government, trade and trend indicators.
- The proposed workflow emphasises source verification, reproducible extraction, preservation of raw data, systematic profiling, cleaning, validation and documentation. The quality framework covers accuracy, completeness, consistency, timeliness, validity, uniqueness, relevance and traceability. By following this strategy, the project will have a stronger foundation for future demand forecasting and overstock/understock risk analysis.

References

- UCI Machine Learning Repository – Online Retail dataset.
 - Kaggle – Public fashion and retail datasets, subject to individual dataset licences.
 - Open Government Data Platform
- India – public datasets and government statistics.
- UN
- Comtrade – international trade statistics.
- Ministry of Textiles, Government of India – textile sector information and reports.
- International Trade Centre (ITC) – trade and market information.
- Google Trends – public search-interest indicators for supplementary trend analysis.
- Relevant annual reports and publicly available publications from recognised textile and retail organisations.